

Junior Makers · Class 4

Smart Dustbin

with Fill-Level Indicator

No touching. No overflowing. Just wave!

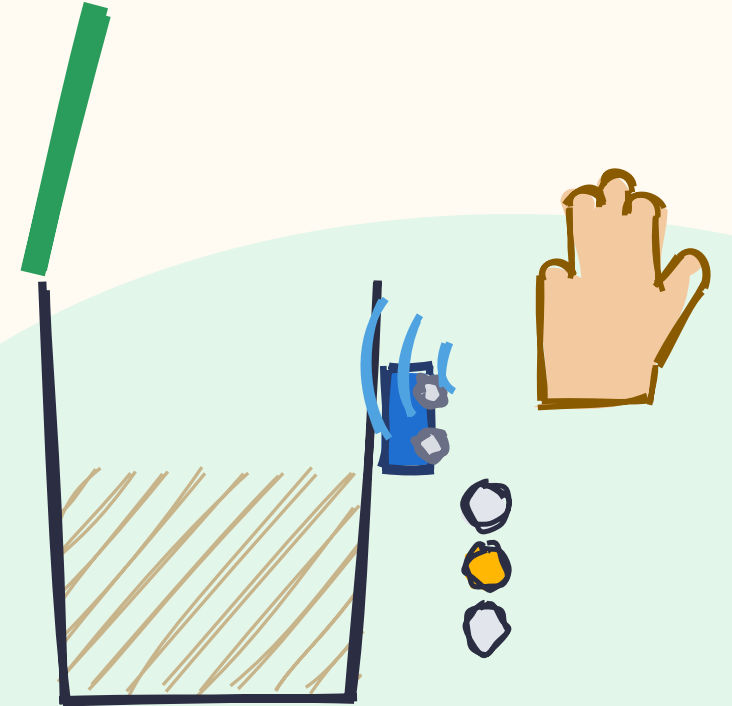
Ultrasonic sensing

Servo motion

Two sensors, one brain

Did you know?

Bats find insects in total darkness using sound. Our bin uses the same trick!





Let's think...

What is wrong with this bin?

Two big problems

1. Everyone touches the same dirty lid. Germs spread from hand to hand.
2. Nobody knows it is full until rubbish falls on the floor.

Open it with your foot?

Check it every hour?

Make the bin smart!



overflowing!

Today: we build a bin that opens when it sees your hand and shows how full it is.





Learning aims

Today I will learn to...



1



Explain how an **ultrasonic sensor** measures distance with echoes

2



Turn a **servo** to an exact angle to open and close a lid

3



Use **two sensors** on one Arduino at the same time

4



Show a **fill level** with a green, yellow and red LED bar

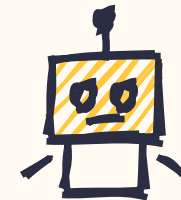
5



Explain how smart bins keep **cities clean**

Keywords

ultrasonic · echo · distance · cm ·
servo · angle · threshold





Real world

Smart bins are already here



Smart city bins

Send a message to the council when they are full, so trucks only go where needed



Hospitals

Touch-free bins stop germs spreading between patients



Fewer truck trips

Less fuel, less traffic, less pollution



Clean and green

No overflowing bins on the streets or in the school



Think: what would happen if every bin in Islamabad could tell the cleaners "I am full, come now"?

Engineer's eye

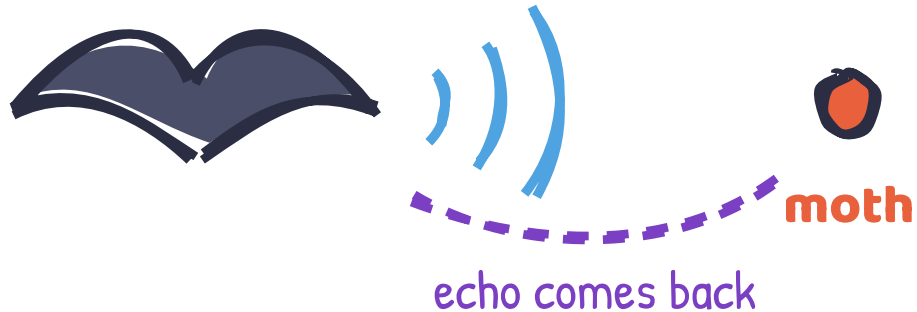
The cleverest machines solve small, everyday problems.





The science

Seeing with sound: echoes



A bat shouts a very high squeak. The sound bounces off a moth and comes back. **Quick echo = close. Slow echo = far.**

Ultrasonic means...

Ultra = beyond. **Sonic** = sound.

A sound too high for humans to hear. Our sensor sends **40,000 vibrations every second** (40 kHz). We can only hear up to about 20 kHz.

Try it now

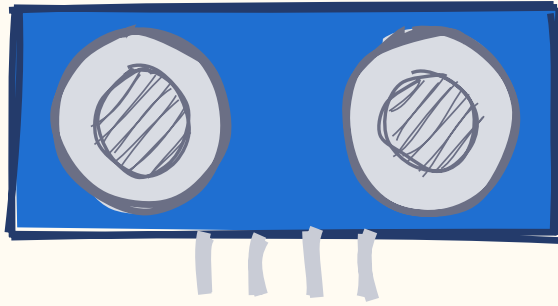
Clap once facing the wall, then from far away. Which echo comes back faster?





Meet the sensor

The HC-SR04 ultrasonic sensor



speaker (sends) microphone (hears)

Fact file

Range: 2 cm to 400 cm

Accuracy: about 3 mm

Beam: a cone about 15° wide

Speed of sound: 343 m/s



The maths inside

Sound goes there **and** back, so:

distance (cm) = echo time (µs) ÷ 58

ArduinoBlocks does this maths for you!

Pin	What it does
VCC	5V power
Trig	Arduino says: SHOUT now!
Echo	Sensor says: I heard it back after ... microseconds
GND	Ground



Stand up and play!

Be the ultrasonic sensor

How to play (pairs)

1. Partner A is the **sensor**, partner B is the **timer**.
2. A rolls a soft ball at the wall and catches it on the way back.
3. B counts how long it takes: "one-elephant, two-elephant..."
4. Move 2 big steps back. Try again!



Far away = **longer** time. The sensor does exactly this, but **millions of times faster**.

Distance

Time

Close (1 m) _____ counts

Middle (3 m) _____ counts

Far (5 m) _____ counts

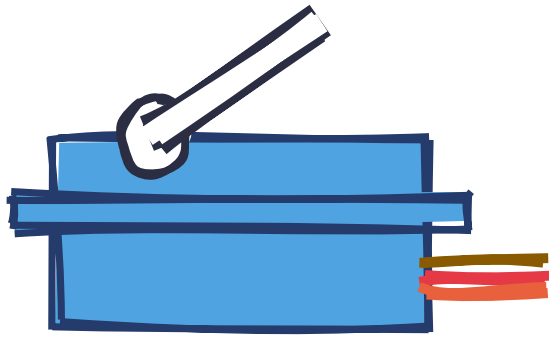
Think deeper

What if the wall were soft, like a curtain? Would the ball (or the sound) bounce back well?

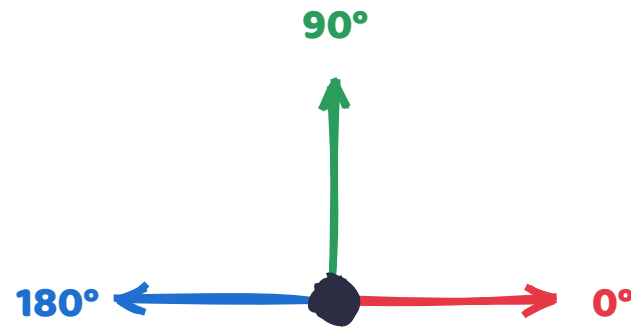


Meet the motor

The SG90 servo: a motor that obeys angles



A normal motor spins and spins. A servo turns to an **exact angle** and **holds it**.



It can turn from 0° to 180°

3 wires

Brown → GND

Red → 5V

Orange → Signal: pin 9



Our lid

0° = lid closed. 90° = lid open. Your bin might need 70° or 110°. Test it!

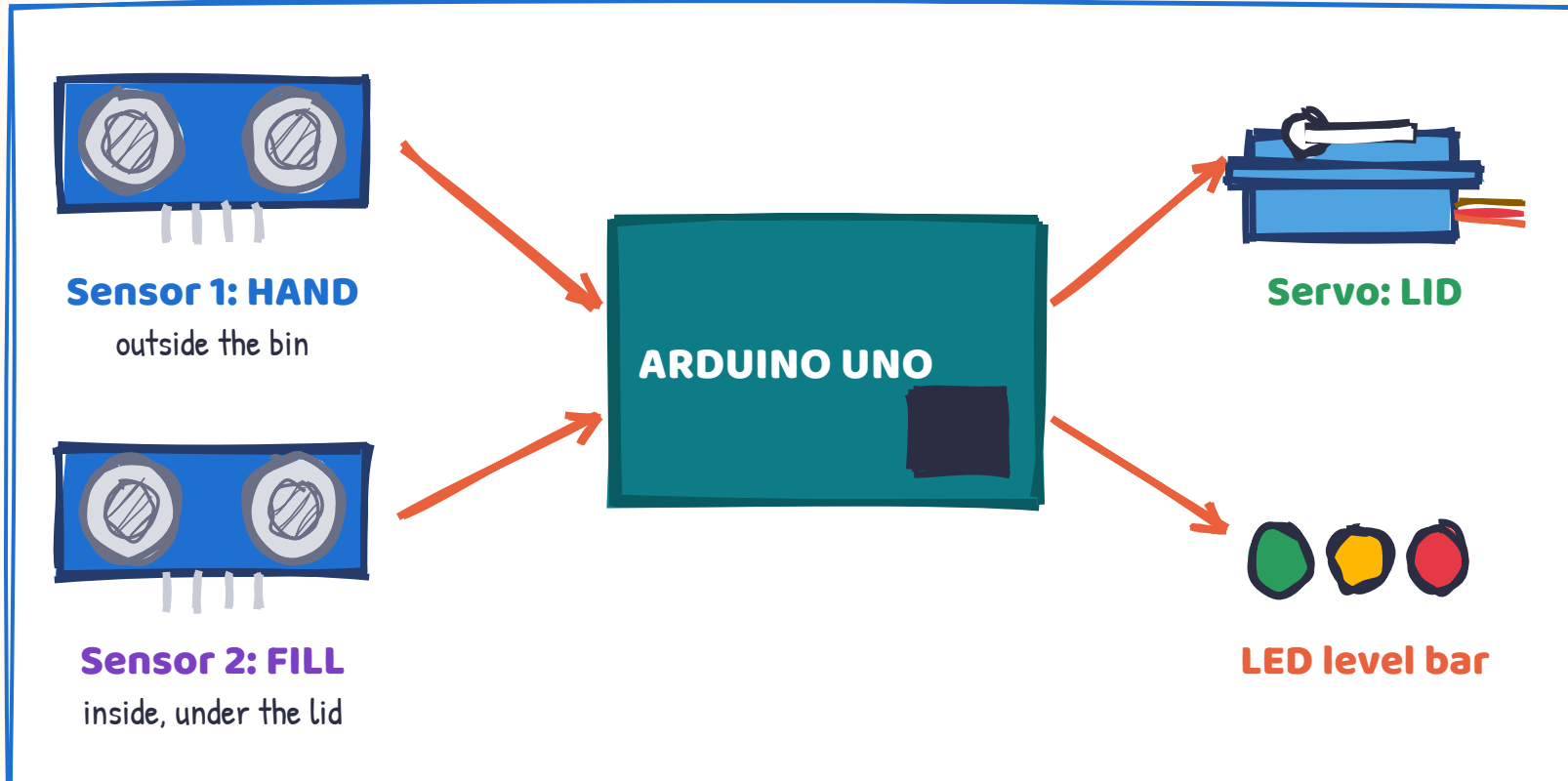
Inside a servo: a small motor, gears to make it strong, and a sensor that checks the angle. It is a tiny robot on its own!





The plan

Two sensors, one brain, two jobs



Job 1: lid

IF a hand is closer than **15 cm**,
open the lid, wait 3 s, close it.

Job 2: level

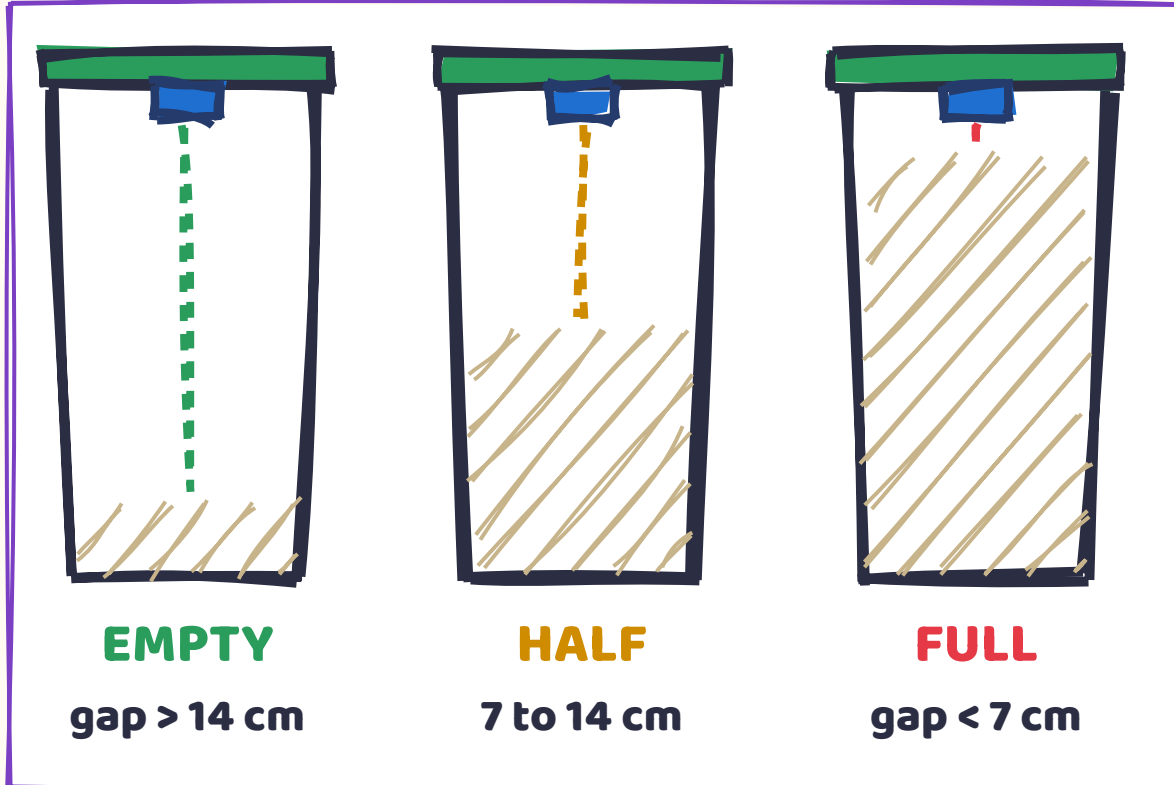
Measure the gap from the lid
down to the rubbish. Show it on
the LEDs.

Input → decision → output. Same pattern,
now doubled!



The clever part

Measuring how full the bin is



Upside-down thinking!

The sensor looks **down** from the top. More rubbish means a **smaller** distance.

Big number = empty. Small number = full.

LED	Gap to rubbish
Green	more than 14 cm
Yellow	7 to 14 cm
Red	less than 7 cm

For a bin 20 cm deep. Measure your own box and split it into thirds.





Build it

Our smart bin kit (per group)

**Arduino Uno**

The brain

**2 × HC-SR04**

Hand + fill sensors

**SG90 servo**

Lifts the lid

**3 LEDs + 3 × 220 Ω**

Green, yellow, red

**Cardboard box**

The bin body + lid

**Jumper wires**

Breadboard + USB

Also: tape, glue gun (teacher only), scissors, a stick or straw for the lid arm, scrap paper "rubbish" to test.

Recycle!

Use an old shoe box or cereal box. Our bin is made from rubbish!

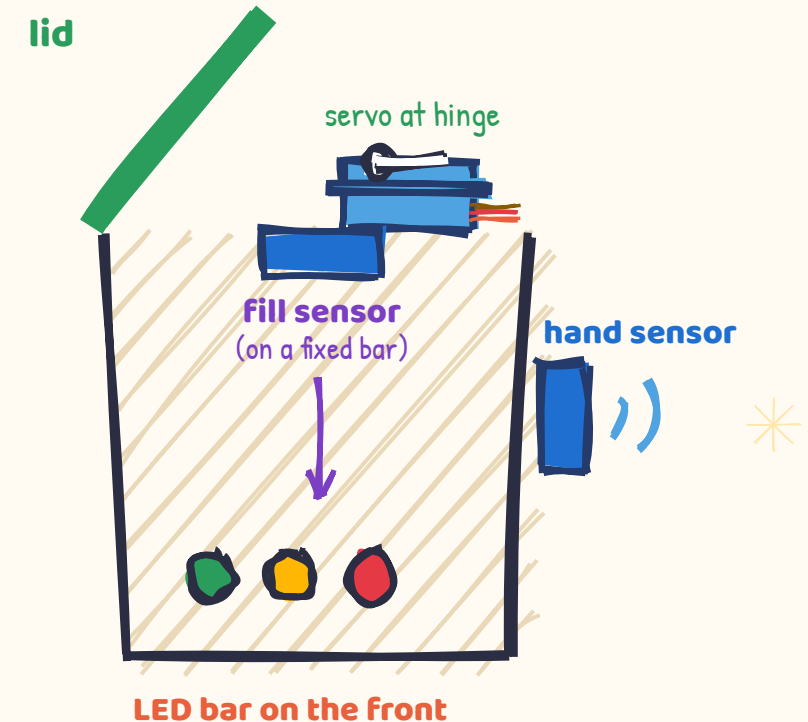




Build it

Make the bin body

- 1 **Cut** a flap lid from the top of the box. Leave one side as the hinge.
- 2 **Fix** the servo next to the hinge. Tape a straw from the horn to the lid.
- 3 **Mount** the fill sensor on a card bar **under** the lid, looking down. Not on the lid!
- 4 **Stick** the hand sensor on the front, looking out.
- 5 **Poke** 3 LEDs through the front: green at the bottom, red at the top.

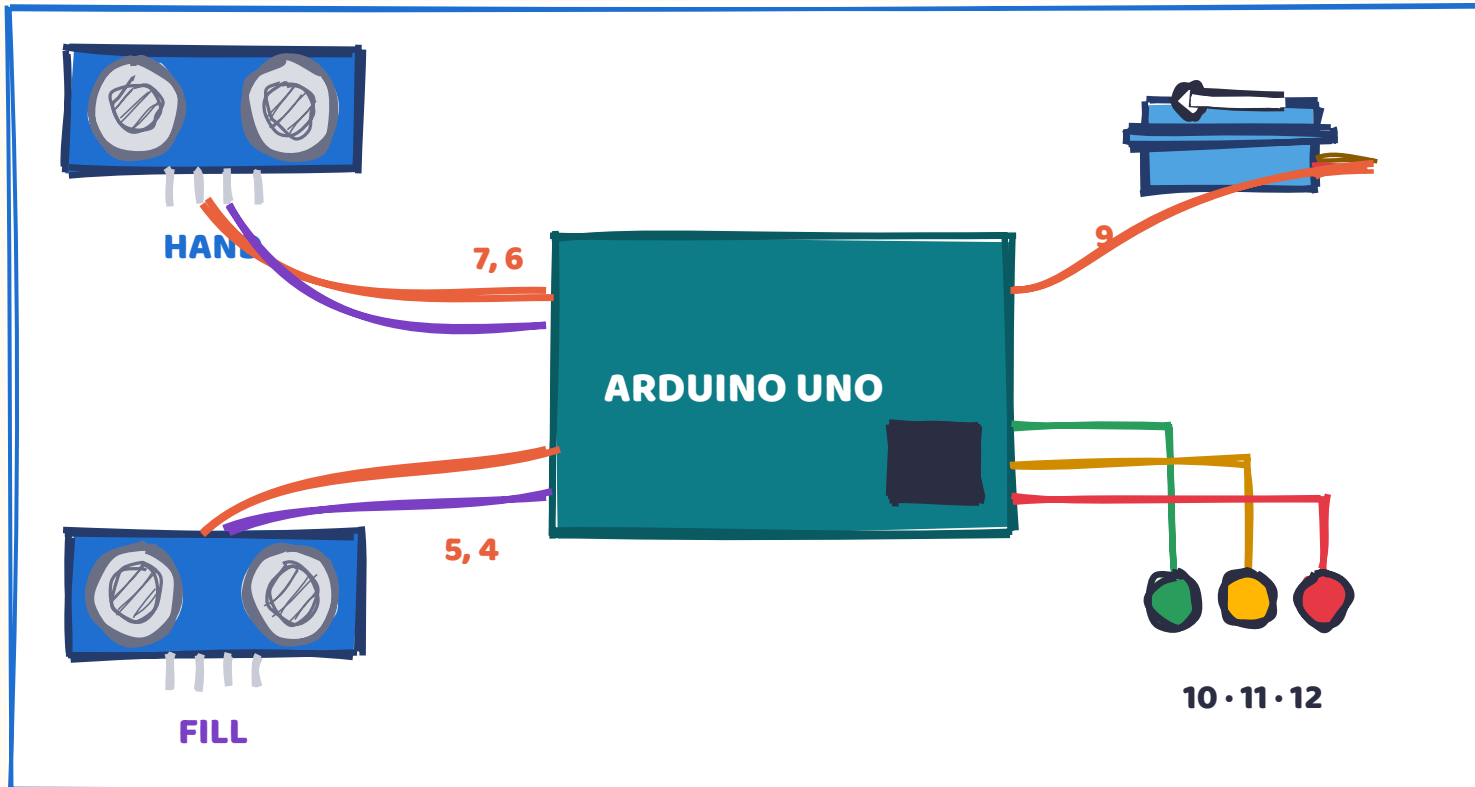


Why not on the lid? When the lid opens, a sensor on it would point at the ceiling!



Build it

Wiring the smart bin



Part	Arduino
Hand Trig / Echo	7 / 6
Fill Trig / Echo	5 / 4
Sensors VCC / GND	5V / GND
Servo orange	pin 9
Servo red / brown	5V / GND
LEDs G / Y / R	10 / 11 / 12
Each LED	+ 220 Ω

Power tip

Servo twitching or Arduino resetting? Plug into a laptop port directly, or use a 5V adapter.



Code it, part 1

Job 1: the hands-free lid

when Arduino starts

servo pin 9 → angle 0 (lid closed)

forever

set hand to ultrasonic distance (Trig 7, Echo 6)

if hand < 15

servo pin 9 → angle 90 (open)

wait 3000 ms

servo pin 9 → angle 0 (close)

wait 1000 ms

Read it like a story

Start with the lid **closed**.

Keep checking for a hand.

IF a hand is closer than 15 cm: open,
wait 3 seconds, close.

Wait 1 second so the lid settles.



Common bug

Lid opens by itself? Something is in front of the sensor, like a wall or your bag. Aim it at open space!



Code it, part 2

Job 2: the fill-level bar



(still inside forever)

set fill to ultrasonic distance (Trig 5, Echo 4)

turn OFF pins 10, 11, 12

if fill < 7

turn ON pin 12 (red: FULL)

else if fill < 14

turn ON pin 11 (yellow: HALF)

else

turn ON pin 10 (green: EMPTY)

wait 200 ms



Ranges, not just yes/no

Last time we used one **IF**. Now we use **IF, ELSE IF, ELSE** to sort a number into **three ranges**.

The order matters: check the smallest number first!



Why one at a time?

Two ultrasonic sensors shouting together can hear each other's echoes. Reading them one after the other keeps them honest.





Test like an engineer

Does every part work?



Test	Expect	Result	Reading
Wave a hand at 10 cm	Lid opens?	pass / fail	_____ cm
Wave at 30 cm	Lid stays shut?	pass / fail	_____ cm
Empty bin	Green?	pass / fail	_____ cm
Half full of paper	Yellow?	pass / fail	_____ cm
Fill to the top	Red?	pass / fail	_____ cm

Debug checklist

Reading stuck at **0**? Check Trig and Echo are not swapped.

Wrong LED? Check the **order** of your IF blocks.

Servo buzzing at 0°? Try **5°** instead.



Open the **Serial monitor** and print both distances. Seeing the numbers makes bugs easy to find!



Level up!

Smart bin challenges

Choose your challenge



Full alarm: add a buzzer that beeps 3 times when the bin turns red.



No entry: when the bin is full, do NOT open the lid. Flash red instead.



Slow close: close the lid gently, a few degrees at a time, with a repeat loop.



Thank you! blink all 3 LEDs as a thank-you after the lid closes.

Design thinking

Who else would love this bin? A doctor? A cook? Someone who cannot bend down? How would you change it for them?

Boss level

Make the lid open only if a hand stays there for half a second. No more accidental opens!



Show what you know

What did we learn?

- ☒ An ultrasonic sensor measures distance using echoes, like a bat
- ☒ Distance (cm) = echo time (μ s) $\div$ 58
- ☒ A servo turns to an exact angle and holds it
- ☒ IF, ELSE IF, ELSE sorts a number into ranges
- ☒ Two sensors + one brain = one smart system

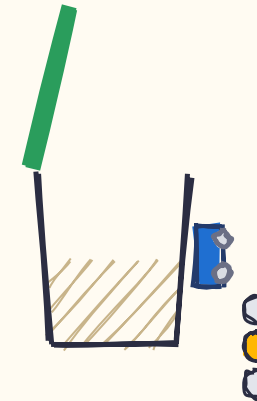


Exit ticket: the fill sensor reads 5 cm. Which LED is on, and why?



At home

Find a machine that opens or switches on by itself when you come near. What sensor might it use?



Next class: a Rain-Sensing Automatic Window!

